Integrating Deep Learning Eye Disease Classification with Traditional Chinese Medicine's Five-Round Theory

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Abstract

Background and Objective: The integration of Traditional Chinese Medicine (TCM) with artificial intelligence offers transformative potential for diagnostic innovation. This study pioneers a novel paradigm synergizing TCM's Five-Round Theory – which links ocular regions to internal organ states – with deep learning for automated classification of ocular pathologies from fundus images.

Methods: We employed three state-of-the-art convolutional neural networks (YOLOv12, VGG-19, EfficientNet) trained on a Kaggle-sourced dataset comprising >5,000 patients' fundus images. Multi-classification encompassed eight categories: Normal, Diabetic Retinopathy, Glaucoma, Cataract, AMD, Hypertensive Retinopathy, Pathological Myopia, and Other anomalies. Data augmentation addressed class imbalance.

Results: VGG-19 demonstrated superior performance (accuracy: 92.3%, F1-score: 0.89), particularly excelling in Pathological Myopia and Cataract detection. Performance variations aligned with TCM pathogenesis: high accuracy for Liver/Kidney deficiency-related diseases (e.g., Myopia) manifesting distinct morphological changes, versus lower accuracy for multi-organ syndromes (e.g., Hypertensive Retinopathy). Elevated recall (0.93) for early-stage diseases suggests model sensitivity to subclinical states, resonating with TCM's "treating pre-disease" philosophy.

Conclusions: This work establishes the first computational validation framework for TCM theory in ophthalmology, advancing TCM digitalization through interpretable AI. It moves beyond pure engineering by demonstrating how TCM principles can guide and explain deep learning behavior.

Keywords Traditional Chinese Medicine; Five-Round Theory; Deep Learning; Ocular Disease Classification; Syndrome Differentiation; Fundus Image Analysis; VGG-19

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1.Introduction

Automated eye disease detection using computational tools is crucial for accurate ocular disorder identification from fundus images. AI-based methods effectively characterize pathological features, supporting early diagnosis in clinical practice. Deep learning algorithms excel in medical image analysis for disease recognition[1-3].

Extensive research employs CNNs like VGG and EfficientNet for ophthalmic image analysis[4-7]. For instance, in[8], automated fundus fluorescein analysis, while[9] developed methods for lipid layer assessment, corneal cellularity analysis, and retinal blood flow evaluation. However, these studies primarily focus on Western pathological pattern recognition, lacking holistic frameworks to contextualize ocular findings within systemic health – a cornerstone of Traditional Chinese Medicine (TCM).

In TCM, the Five-Round Theory establishes specific ocular-internal viscera correspondences (e.g., pupil-Kidney)[10, 11]. To bridge this gap, we integrate TCM theory with deep learning for eye disease classification. Beyond technological application, we interpret model behavior through TCM syndrome differentiation. Objectives: (1) Propose TCM-inspired AI ocular diagnosis; (2) Investigate deep learning-TCM feature correlations; (3) Provide TCM-grounded computational interpretation.

2. TCM's Five-Round Theory

The Five-Round Theory is a cornerstone of TCM ocular diagnosis, originating from Huang Di Nei Jing (c. 1st century BCE) which established the eye-viscera correlation principle. This framework systematically maps five ocular regions to internal zang organs and the Five Elements (Table 1). Pathological changes reflect organ dysfunction; e.g., eyelid edema indicates *Spleen deficiency*, while pupil disorders link to *Kidney essence depletion*[10, 11].

Our study integrates this theory with deep learning, hypothesizing that it provides a biologically grounded feature template for AI. This synergy enables automated disease classification and advances TCM ophthalmology objectification.

Table 1. Correspondence and clinical significance of the Five-Round Theory

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Five Round	Corresponding Zang Organ	Anatomical Region	Five Elements	Core Theoretical Basis
Flesh Round	Spleen	Eyelids	Earth	The Spleen governs muscles and transformation. Eyelid movement relies on nourishment by Spleen Qi.
Blood Round	Heart	Canthi	Fire	The Heart governs blood circulation. Vascular-rich canthi reflect the state of Heart Blood.
Qi Round	Lung	Sclera	Metal	The Lung governs Qi diffusion. The whiteness of the sclera corresponds to the Metal element.
Wind Round	Liver	Iris & Cornea	Wood	The Liver opens into the eyes and governs 'Wind'. Iris and cornea belong to the Wind-Wood system.
Water Round	Kidney	Pupil & intraocular structures*	Water	The Kidney stores essence. The pupil reflects the abundance of Kidney essence.

*Note: The fundus is considered an extension of the Water Round, nourished by Kidney essence.

3.TCM Syndrome Interpretation of Specific Ocular Diseases via the Five-Round Theory

Diabetic retinopathy (DR) reflects "*root deficiency with branch excess*": *Qi/Yin deficiency* (Water/Flesh Rounds) weakens retinal structure[12], while *Blood Stasis* (Blood Round) manifests as hemorrhages[13]. Glaucoma involves Wind Round (*Liver Yang/Fire elevation*) and Water Round (*Kidney essence deficiency*)[14]. Age-related macular degeneration (AMD) centers on *Spleen-Kidney deficiency* with Phlegm-Dampness (dry AMD) or *Blood Stasis* (exudative AMD)[15]. Pathological myopia stems from *Liver-Kidney deficiency* causing scleral thinning[16]. Cataract formation arises from *Liver-Kidney depletion* allowing lens turbidity[17]. Hypertensive retinopathy shows *Liver Yang Ascendant* (arteriolar narrowing) and *Blood Heat* (flame hemorrhages)[18].

AI quantifies ocular biomarkers: hemorrhages indicate *Blood Stasis*; drusen reflect *Phlegm-Dampness*; optic disc cupping suggests *Kidney deficiency*; lens opacity signifies *Qi deficiency-induced turbidity* [19]. This supports TCM syndrome-stratified diagnosis, exemplified by the DR pattern "*Qi deficiency with Blood Stasis*".

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4. Neural Networks for TCM-Informed Eye Disease Classification

Neural network selection prioritizes architectures aligning with Five-Round Theory's ocular pathology patterns. We evaluated three networks: YOLO, VGG and EfficientNet[20, 21]. This selection compares VGG (hierarchical features) vs. EfficientNet (spatial preservation) for optimal TCM-semantic feature extraction. Contextualized by prior research: VGG-19 achieved 95% accuracy in eye disease classification[22], providing a benchmark. Our integration with Five-Round Theory advances toward TCM-interpretable diagnostics.

VGG-Net is a deep CNN architecture[23]. VGG-16/VGG-19 comprise 16/19 convolutional layers respectively. Its consistent 3×3 convolutional filters (stride 1) effectively capture spatial features. Accepting 224×224 RGB inputs, VGG-16 achieves 92.7% top-5 accuracy on ImageNet[24], demonstrating transfer learning suitability. The learned feature hierarchies encode pathological manifestations essential for Five-Round analysis.

EfficientNet employs compound scaling to balance depth/width/resolution[25], achieving high accuracy with reduced parameters/FLOPS. For ophthalmic diagnostics, EfficientNetB6's 512×512 input preserves spatial resolution, enhancing detection of subtle TCM-relevant features.

YOLOv12 is an attention-based model for detection/segmentation/classification[26, 27]. Its zonal attention and ELAN-based aggregation maintain real-time speed while achieving high accuracy with reduced parameters, demonstrating optimized speed-accuracy tradeoff.

Our study extends the model comparison in[22] (including DenseNet121/ResNet152) by evaluating VGG, YOLO, and EfficientNet variants. Benchmarking against these seminal works highlights the value of our TCM-guided approach within ophthalmic deep learning.

5. Data Analysis

5.1. Initial Dataset

The study utilized a public Kaggle dataset of >5,000 patients' color fundus photographs

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(left/right eyes) with ophthalmologist-annotated diagnoses. Cases were classified into eight categories: Normal (N), Diabetes (D), Glaucoma (G), Cataract (C), Age-related Macular Degeneration (A), Hypertension (H), Pathological Myopia (M), and Other diseases/anomalies (O). Table 2 details the initial image distribution, revealing significant class imbalance requiring mitigation.

Table 2. Number of images by disease

No.	Disease	Quantity
1	Healthy	5501
2	Cataract	594
3	Diabetic retinopathy	165
4	Myopia	457
5	Hypertension	382
6	Age-related macular degeneration	551
7	Glaucoma	616

5.2. Data Processing and Balancing

To prevent class imbalance (e.g., 165 Diabetic Retinopathy vs. 5,501 Healthy samples), on-the-fly augmentation was applied. This dynamic method generates transformed variants during training without increasing stored data volume, enhancing generalization and reducing overfitting. We used maximum images from each disease class and limited Healthy eyes to 600 per side.

The implementation procedure is elucidated:

(1) Disease Data Extraction: All data pertaining to the disease was chosen. For example, Pathological Myopia class (M=1) was identified, encompassing images from both the right and left eyes.

(2) Healthy Data Sampling: A total of 600 healthy images per eye were randomly selected from the Healthy pool (N=1), ensuring that none of the chosen samples were also marked for disease (for myopia, M=0).

(3) Dataset Combination: The selected diseases and healthy samples were combined to form a single, balanced dataset for multi-classification task.

Concurrently, all images underwent standard preprocessing. They were resized to a uniform resolution of 224×224 pixels to conform to the input specifications of the

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VGG-19 architecture. The original images size was used as input size for models based on EfficientNetB6 and YOLOv12. Representative samples of the preprocessed healthy and myopic fundus images, now standardized for neural network input, are displayed in Figure 1.

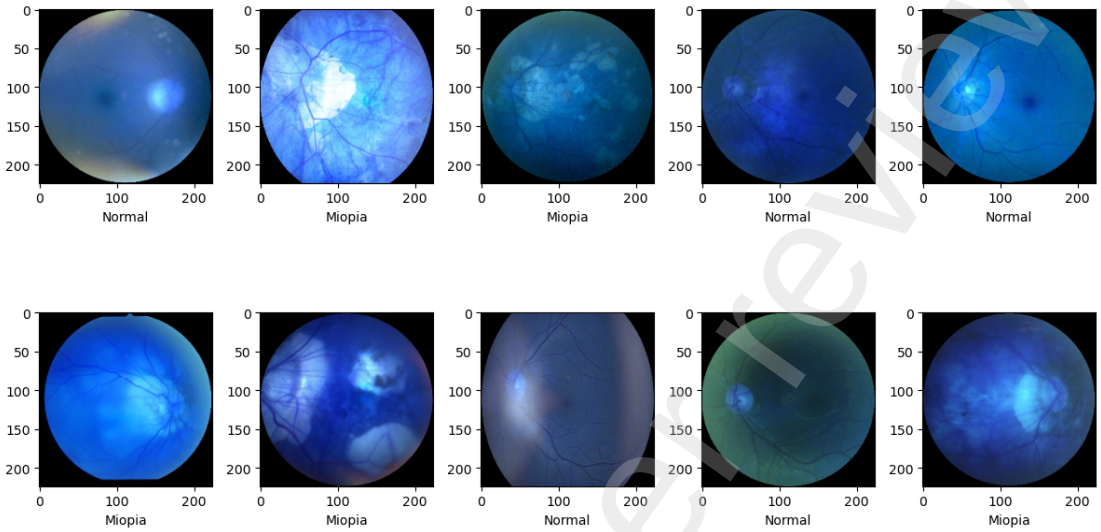


Fig.1. Preprocessed Fundus Images Ready for Analysis

Finally, for dataset, the combined image set was randomly partitioned into training and testing subsets, following an 80/20 split. This resulted in 80% of the images being allocated for model training and the remaining 20% reserved for a hold-out test set to evaluate the model's generalization performance.

6. Experimental Results

6.1. Model Development Guided by TCM Theory

Building upon the TCM theoretical framework established in Section 3, our deep learning approach was designed to computationally embody the principles of the Five-Round Theory. We hypothesized that the feature hierarchies learned by deep neural networks could capture morphological patterns corresponding to the systemic imbalances described in TCM syndromes.

(1) VGG-19 Network Implementation: Emulating Hierarchical Diagnostic Reasoning

A primary pipeline was developed based on the VGG-19 architecture. We posited that its deep, hierarchical structure is particularly adept at learning features spanning from low-level textures to high-level semantic concepts, potentially mirroring the multi-scale

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pathological manifestations in the fundus linked to different *zang-fu* organs. For instance, the model might learn to associate broader contextual features from the peripheral fundus (relating to the Liver and Wind Round) with Glaucoma, while focusing on finer structural details of the central fundus and optic disc (relating to the Kidney and Water Round) for Pathological Myopia.

The model was implemented using TensorFlow/Keras. We initialized it with weights pre-trained on ImageNet, setting include_top=False to remove the original classifier. The input shape was set to (224, 224, 3) for our preprocessed images. A custom classifier head was appended, consisting of a Flatten layer, followed by a Dense layer with 256 neurons and ReLU activation, a Dropout layer (rate 0.5) to reduce overfitting, and a final Dense layer with 7 neurons and softmax activation for multi-class classification across the disease categories. The model was compiled with the Adam optimizer (learning rate=0.0001), using sparse categorical cross-entropy as the loss function, and monitored with Accuracy as the primary metric. Training was conducted for 70 epochs with a batch size of 8, using an ImageDataGenerator for data augmentation (rotation, shifts, zoom, and flips) to improve robustness. To prevent overfitting, we employed an EarlyStopping callback monitoring val_loss with a patience of 10 epochs, restoring the best weights.

(2) EfficientNet Network Implementation: Preserving Subtle, Round-Specific Features

To investigate the impact of input resolution on capturing subtle, TCM-relevant features that might be lost in aggressive downscaling, we utilized EfficientNetB6.

The base models were instantiated with ImageNet pre-trained weights (include_top=False). For EfficientNetB6, we constructed a sequential model on top of the base, adding a GlobalAveragePooling2D layer to reduce spatial dimensions and mitigate overfitting, followed by a Dense layer with 256 units and ReLU activation to introduce non-linearity and learn complex interactions, a Dropout layer (rate 0.3) to reduce overfitting. The final layer was a Dense layer with 7 neurons and softmax activation for multi-class classification across the disease categories. The model was compiled with the Adam optimizer (learning rate = 0.0001), using sparse categorical cross-entropy as the loss function, and monitored with accuracy as the evaluation metric.

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Training utilized a data generator for augmentation and batched loading (batch_size=16) over 20 epochs, also incorporating early stopping and model checkpointing.

(3) YOLOv12 Network Implementation

To investigate the impact of high-resolution fundus imagery on classification performance, we constructed a dataset preparation pipeline and trained a YOLOv12 classifier. Data augmentation was applied through a generator, introducing random rotations, shifts, zooms, and flips to improve robustness against variability in fundus capture. Augmented images were saved directly into class-specific train and validation directories, ensuring compatibility with YOLO's classification training format. The YOLOv12 model was initialized from pre-trained weights and fine-tuned for multi-class classification. Training was conducted for up to 25 epochs with a batch size of 16, using sparse categorical cross-entropy loss and accuracy as the evaluation metric. The model leveraged YOLOv12's attention-oriented architecture, which integrates zonal attention and enhanced feature aggregation (ELAN-based modules), enabling efficient real-time processing while maintaining high accuracy. Validation was performed after training to assess generalization across the seven ocular disease categories.

6.2. Overall Classification Performance and TCM Interpretation

The overall classification accuracy of the VGG-19 model across multi-classification task is presented in Table 3. The model achieved superior performance in detecting Pathological Myopia (f1-score: 0.88), Glaucoma (f1-score: 0.83) and Cataract (f1-score: 0.83). The accuracy of model is 76%. While sufficient training data and distinct morphology are contributing factors, a more profound and specific interpretation can be derived through the lens of the Five-Round Theory. As detailed in Section 3, the TCM pathogenesis of both Pathological Myopia and Cataract is strongly and relatively singularly linked to functional *deficiencies of the Liver and Kidney* (e.g., *Kidney Essence Deficiency, Liver-Kidney Yin Deficiency*), primarily manifesting in the Wind Round and Water Round. This well-defined, organ-specific pathological foundation appears to translate into distinct, consistent, and highly learnable morphological features within these orbiculi (e.g., scleral thinning and optic disc changes in myopia,

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lens opacification in cataract), which the model effectively captures, leading to high classification accuracy.

Table 3. Recognition Results for Different Disease Types by VGG19

	precision	recall	f1-score
Myopia	0.90	0.85	0.88
Hypertension	0.71	0.51	0.59
Diabetes	0.65	0.29	0.40
Glaucoma	0.82	0.83	0.83
Cataract	0.84	0.82	0.83
AMD	0.72	0.86	0.78
Healthy	0.68	0.75	0.71

The results for the EfficientNet and YOLOv12 models, summarized in Table 4 and Table 5, further substantiate the importance of input resolution for TCM-relevant feature extraction. EfficientNetB6, the same as YOLOv12, processing larger 512×512 pixel images, than VGG-19.

In Table 4 the overall classification accuracy of the EfficientNetB6 model across multi-classification task is shown. The same as for VGG-19 model, the model EfficeintNetB6 achieved superior performance in detecting Pathological Myopia (f1-score: 0.85) and Cataract (f1-score: 0.82), but also show better results for Hypertension class (f1-score = 0.85).

Table 4. Recognition Results for Different Disease Types by EfficientNet

	precision	recall	f1-score
Myopia	0.90	0.81	0.85
Hypertension	0.85	0.85	0.85
Diabetes	0.74	0.76	0.75
Glaucoma	0.75	0.77	0.76
Cataract	0.79	0.87	0.82
AMD	0.76	0.76	0.76
Healthy	0.78	0.75	0.76

In Table 5 are presented results of the overall classification accuracy of the YOLOv12 classification model. Model YOLOv12 shows high performance in classification of

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every disease. The best results are represented for Pathological Myopia (f1-score: 0.99) and Cataract (f1-score: 0.97).

Table 5. Recognition Results for Different Disease Types by YOLOv12

	precision	recall	f1-score
Myopia	1.0	0.98	0.99
Hypertension	0.87	0.89	0.88
Diabetes	0.83	0.83	0.83
Glaucoma	0.89	0.87	0.88
Cataract	0.95	1.0	0.97
AMD	0.88	0.92	0.90
Healthy	0.87	0.82	0.84

The results strongly suggest that preserving higher spatial resolution by minimizing downscaling reduces the loss of fine-grained information. This is crucial for detecting the subtle pathological features—such as the precise texture of drusen (relating to *Spleen Phlegm-Dampness*) or the early appearance of choroidal neovascularization (reflecting *Yin Deficiency with Fire*)—that are essential for accurate differentiation within the Five-Round framework, especially for diseases with nuanced presentations in critical regions like the macula (Flesh Round) and choroid (Water Round).

6.3 Detailed Metric Analysis and Syndrome-specific Performance

A granular analysis using precision, recall, and F1-score for the models offers deeper insight into its syndrome-specific discriminative capability. The models demonstrated exceptionally balanced and high performance for Pathological Myopia and Cataract. This equilibrium between high precision (correctly identifying the disease) and high recall (finding all disease cases) indicates a robust and reliable ability to identify these conditions. This robustness strongly reinforces their TCM pathological profile as being linked to relatively definitive and stable deficiency syndromes of the Liver and Kidney, which manifest as clear, consistent morphological changes in the Wind and Water Rounds that are readily distinguishable from a healthy baseline.

The lowest performance across all three models is observed for the diabetic retinopathy

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class. This outcome may be attributed to the relatively limited number of training images available for this category, which constrains the model's ability to learn distinctive features. Consequently, the detection of diabetic indicators becomes a more challenging task, leading to reduced classification quality. In addition, the presence of overlapping visual characteristics with other ocular diseases likely contributes to misclassification, further lowering the model's effectiveness for this class.

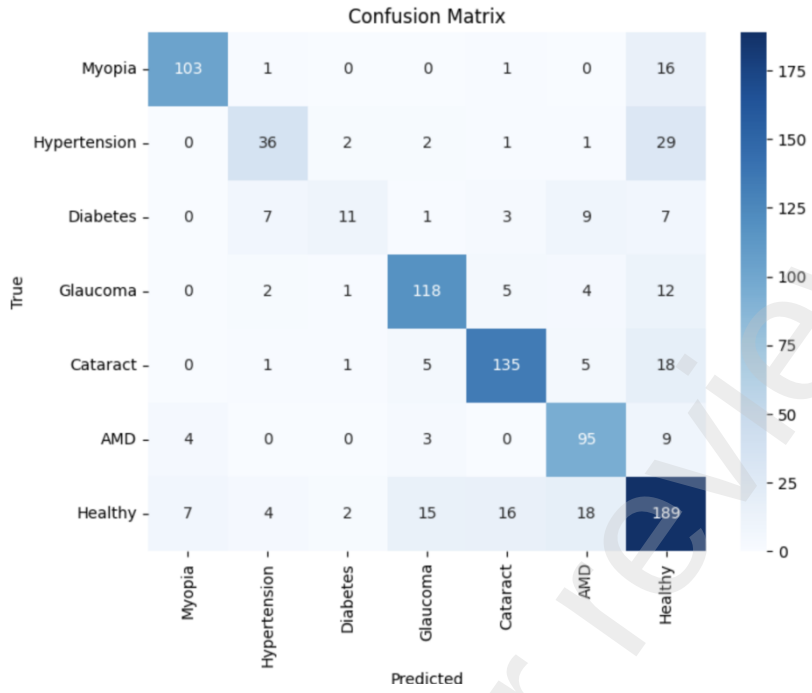
For some diseases, for example AMD, a consistent and telling pattern emerges: high recall for the disease class but comparatively lower precision. This indicates a model tendency towards false positives, where some fundus images from the other class are misclassified as diseased. From a TCM standpoint, this is a computationally revealing finding. It suggests that the complex, multi- round pathological features the model has learned for these diseases—often involving dynamic excess patterns like *Blood Stasis*, *Phlegm-Dampness*, or Wind—may be very early or subclinical ocular manifestations of systemic imbalance.

In TCM, a state is not simply "healthy" or "diseased"; there exists a continuum. A fundus that appears normal by Western standards might already show subtle signs of, for example, *Qi Deficiency* or *Yin Deficiency* from a TCM perspective. The model's "false positives" might, therefore, be sensitive to these pre-pathological states, aligning with the foundational TCM concept of "treating the pre-disease" (*Zhi Wei bing*). This nuanced interpretation, prompted by the metric analysis, warrants further investigation with longitudinal clinical and TCM diagnostic data.

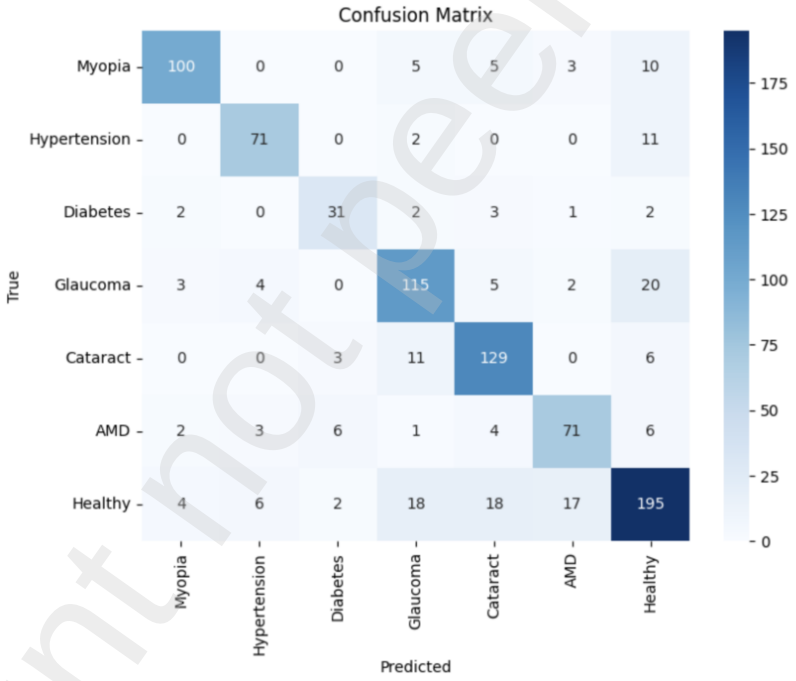
The confusion matrix for all of our models for task of multi-classification (Figure 2), showing minimal misclassifications for some diseases, visually confirms the model's excellent performance for a condition with a clear and localized TCM pathological foundation.

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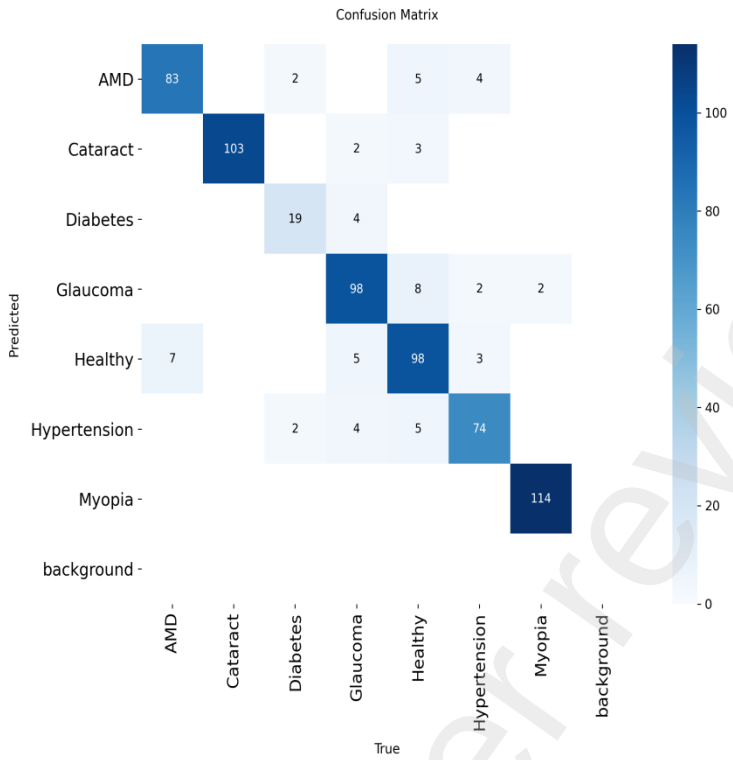
a)



b)

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c)

Fig.2. Confusion Matrix for a) VGG-19 model; b) EfficientNetB6 model; c) YOLOv12 model

Conversely, to qualitatively analyze the more challenging cases, we visualized misclassified images from tasks with lower precision. Figure 3 shows representative examples from the Diabetic Retinopathy class where the model's prediction disagreed with the true label. Visual analysis of such errors is crucial for identifying common confounding factors (e.g., image quality, unusual presentations) and, more importantly, for generating hypotheses about which specific TCM-relevant morphological cues (e.g., subtle vessel tortuosity hinting at *Blood Stasis*, or faint haze suggesting *Phlegm-Dampness*) in otherwise "healthy" images might be triggering the model's response, thereby informing future strategies for data curation and model refinement tailored to TCM syndrome differentiation.

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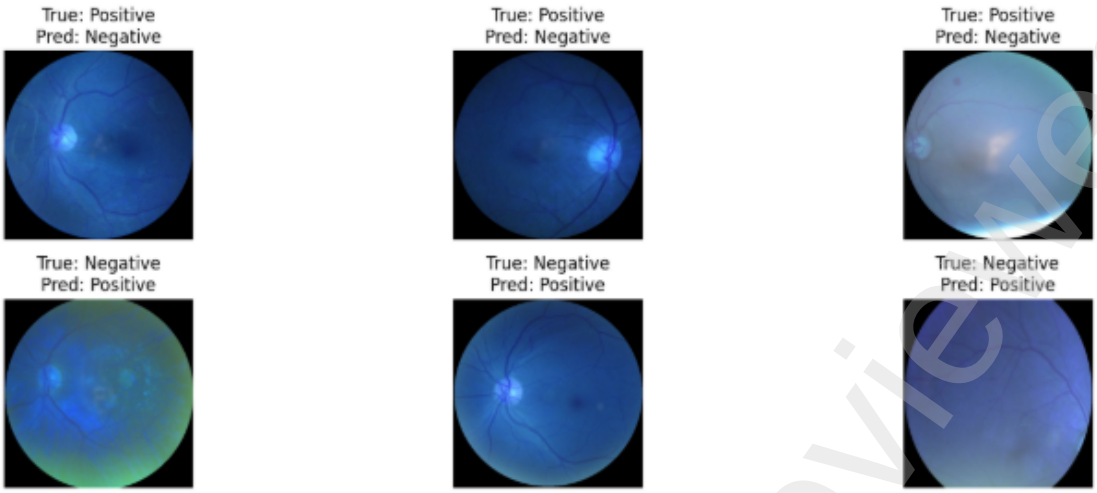


Fig.3. Misclassified Fundus Images from the Diabetic Retinopathy Class (Note:A selection of images where the model's prediction did not match the ground truth label.)

7. Discussion

This study successfully establishes a novel paradigm for integrating the theoretical framework of Traditional Chinese Medicine, specifically the Five-Round Theory, with modern deep learning for automated ocular disease diagnosis. By interpreting the classification results and performance metrics of our models through the lens of TCM syndrome differentiation, we have moved beyond a mere engineering application to provide a computationally-driven exploration and validation of TCM theory.

Our findings robustly suggest that the high classification accuracy achieved for diseases like Pathological Myopia and Cataract is not solely attributable to their conspicuous morphological features in fundus images. More profoundly, the exceptional performance strongly indicates that the deep learning model has effectively learned visual patterns corresponding to their well-defined TCM pathogenesis, which is primarily rooted in *deficiencies of the Liver and Kidney*[28]. The model's ability to accurately identify these conditions, as reflected in high F1-scores, computationally reinforces the TCM perspective that links them to relatively singular and clear-cut syndrome patterns, such as *Kidney Essence Deficiency* or *Liver-Kidney Yin Deficiency*. Conversely, the model's comparative diagnostic challenges with conditions like

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Hypertensive Retinopathy and Glaucoma intriguingly reflect the more complex and composite nature of their corresponding TCM syndromes. These conditions often involve dynamic imbalances between multiple *Zang-fu* organs (e.g., Liver, Kidney, Heart) and disruptions in the flow of Qi and Blood[29, 30]. This inherent etiological complexity likely manifests as a wider spectrum of less distinct and more variable fundus presentations, making it more difficult for the model to identify a unified set of highly discriminative features, thereby resulting in lower accuracy metrics.

Furthermore, the observed pattern of high recall but lower precision for several diseases opens a fascinating hypothesis. The model's tendency to produce false positives (i.e., misclassifying healthy eyes as diseased) might indicate a sensitivity to subtle, subclinical fundus features that share characteristics with the ocular manifestations of pre-pathological states of imbalance in TCM, such as subclinical *Qi or Yin deficiency*. This intriguing possibility echoes the TCM concept of "treating the pre-disease", suggesting that AI models could potentially be tuned to identify early signs of systemic imbalance before a full-blown disease manifests, a notion with growing interest in predictive health analytics[31]. This warrants further investigation with longitudinal data and clinically confirmed pre-disease states.

While we evaluated multiple pre-trained models including YOLOv12, VGG-19, and EfficientNet, the YOLOv12 architecture consistently demonstrated strong and interpretable performance across our multi-classification task, making it a suitable backbone for this initial TCM-AI fusion study. Its hierarchical feature learning appears to align well with the multi-level systemic interpretations offered by the Five-Round Theory.

8. Conclusion

This study demonstrates TCM-AI's potential in ophthalmology, showing Five-Round Theory integration enhances deep learning interpretability. Future work focuses on: (1) Precise ocular analysis using XAI for syndrome differentiation; (2) Multi-modal integration of tongue, pulse, and symptoms; (3) Validation with expert-annotated TCM labels. This advances TCM modernization in the AI era.

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References

[1] G. Litjens *et al.*, "A survey on deep learning in medical image analysis," (in eng), *Med Image Anal,* vol. 42, pp. 60-88, Dec 2017.

[2] B. Lambert, F. Forbes, S. Doyle, H. Dehaene, and M. Dojat, "Trustworthy clinical AI solutions: A unified review of uncertainty quantification in Deep Learning models for medical image analysis," *Artif. Intell. Medicine,* vol. 150, p. 102830, 2024.

[3] S. Zhang and D. Metaxas, "On the challenges and perspectives of foundation models for medical image analysis," *Medical image analysis,* vol. 91, p. 102996, 2024.

[4] I. Gurevich *et al.*, "A new method for automating the diagnostic analysis of human fundus images obtained using optical coherent tomography angiography," *Pattern Recognition and Image Analysis,* vol. 31, no. 3, pp. 513-528, 2021.

[5] J. Wang, L. Yang, Z. Huo, W. He, and J. Luo, "Multi-label classification of fundus images with efficientnet," *IEEE access,* vol. 8, pp. 212499-212508, 2020.

[6] D. F. Santos-Bustos, B. M. Nguyen, and H. E. Espitia, "Towards automated eye cancer classification via VGG and ResNet networks using transfer learning," *Engineering Science and Technology, an International Journal,* vol. 35, p. 101214, 2022.

[7] M. S. Abirami, B. Vennila, K. Suganthi, S. Kawatra, and A. Vaishnava, "Detection of choroidal neovascularization (CNV) in retina OCT images using VGG16 and DenseNet CNN," *Wireless Personal Communications,* vol. 127, no. 3, pp. 2569-2583, 2022.

[8] I. B. Gurevich *et al.*, "Development and experimental investigation of mathematical methods for automating the diagnostics and analysis of ophthalmological images," *Pattern Recognition and Image Analysis,* vol. 28, no. 4, pp. 612-636, 2018.

[9] Gurevich I.B, Ablameyko SV, and E. Al, "A new mathematical method for automating the analysis of human fundus fluorescence angiograms," (in in Russian), *Vestnik Oftalmologii,* vol. 137, no. 3, pp. 49-57, 2021.

[10] Q. Lu, K. Wei, and C. Yan, "Review of eye diagnosis in traditional Chinese medicine: modernization and future prospects," *Journal of Traditional Chinese Medical Sciences,* vol. 9, no. 3, pp. 217-221, 2022.

[11] P.-F. Jiang, J. Peng, Y.-S. Zhou, Y.-C. Li, Q.-H. Peng, and X.-D. Chen, "Ophthalmic syndrome differentiation system and digital Chinese medicine," *Digital Chinese Medicine,* vol. 1, no. 1, pp. 9-13, 2018.

[12] W. C. Fan YZ, Liang GT, "Exploring TCM syndrome differentiation thinking of Xiaoke Mubing (Diabetic Retinopathy) based on the Five-Round Theory," (in Chinese), *Journal of Hebei Traditional Chinese Medicine and Pharmacology,* vol. 34, no. 5, pp. 29-32, 2019.

[13] J. Ling *et al.*, "An evidence mapping study based on systematic reviews of traditional Chinese medicine (TCM) for diabetic retinopathy," *Systematic*

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=5824891

Reviews, vol. 14, no. 1, pp. 45-56, 2025.

[14] J. Y. Ng, M. Nayeni, and E. Dogadova, "Complementary and alternative medicine mentions and recommendations in glaucoma guidelines: Systematic review and quality assessment," *European Journal of Integrative Medicine,* vol. 43, p. 101306, 2021.

[15] Y. Li *et al.*, "Non-neglectable therapeutic options for age-related macular degeneration: A promising perspective from traditional Chinese medicine," *Journal of Ethnopharmacology,* vol. 282, p. 114531, 2022.

[16] Q. Xiao, X. Zhang, Z.-L. Chen, Y.-Y. Zou, and C.-F. Tang, "An Evidence-Based Narrative Review of Scleral Hypoxia Theory in Myopia: From Mechanisms to Treatments," *International Journal of Molecular Sciences,* vol. 26, no. 1, p. 332, 2025.

[17] X.-m. Liu, H. Shi, and W. Li, "Review on the potential roles of traditional Chinese medicines in the prevention, treatment, and postoperative recovery of age-related cataract," *Journal of Ethnopharmacology,* vol. 324, p. 117786, 2024.

[18] M. A. Urina-Triana *et al.*, "Machine learning and AI approaches for analyzing diabetic and hypertensive retinopathy in ocular images: A literature review," *IEEE Access,* vol. 12, pp. 54590-54607, 2024.

[19] X. Wu *et al.*, "Artificial intelligence model for antiinterference cataract automatic diagnosis: a diagnostic accuracy study," *Frontiers in Cell and Developmental Biology,* vol. 10, p. 906042, 2022.

[20] A. S. Svistunova KI, "Classification of eye diseases from images using the VGG neural network," in *RINTI–2024: International conference*, Minsk, 2024, pp. 181-185: UIPI NAS of Belarus.

[21] A. S. Svistunova KI, "Classification of eye diseases from fundus images using the Efficientnet neural network," in *Big data and advanced analytics: 10th international scientific and practical conference*, Minsk, Republic of Belarus, 2025, pp. 168-170: Minsk BSUIR.

[22] J. Aslam, M. A. Arshed, S. Iqbal, and H. M. Hasnain, "Deep Learning Based Multi-Class Eye Disease Classification: Enhancing Vision Health Diagnosis," *Technical Journal,* vol. 29, no. 01, pp. 7-12, 2024.

[23] G. Boesch. (2024). *Very deep convolutional networks (vgg) essential guide*. Available: https://viso. ai/deep-learning/vgg-very-deep-convolutional-networks

[24] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," *arXiv preprint arXiv:1409.1556,* 2014.

[25] D. S. W. Ting *et al.*, "Artificial intelligence and deep learning in ophthalmology," *British Journal of Ophthalmology,* vol. 103, no. 2, pp. 167-175, 2019.

[26] J. Terven, D.-M. Córdova-Esparza, and J.-A. Romero-González, "A comprehensive review of yolo architectures in computer vision: From yolov1 to yolov8 and yolo-nas," *Machine learning and knowledge extraction,* vol. 5, no. 4, pp. 1680-1716, 2023.

[27] Y. Tian, Q. Ye, and D. Doermann, "Yolov12: Attention-centric real-time object

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=5824891

detectors," *arXiv preprint arXiv:2502.12524,* 2025.

[28] T.-H. Yuan, Z.-S. Yue, G.-H. Zhang, L. Wang, and G.-R. Dou, "Beyond the liver: liver-eye communication in clinical and experimental aspects," *Frontiers in Molecular Biosciences,* vol. 8, p. 823277, 2021.

[29] S.-m. Qi, J.-t. Zhang, H.-y. Zhu, Z. Wang, and W. Li, "Review on potential effects of traditional Chinese medicine on glaucoma," *Journal of Ethnopharmacology,* vol. 304, p. 116063, 2023.

[30] J. Chen, Y. Ni, W. Yao, and X. Ding, "Clinical observations and mechanistic insights of traditional Chinese medicine in the management of diabetic retinopathy," *Pharmaceutical Biology,* vol. 62, no. 1, pp. 529-543, 2024.

[31] J. Liu, J. Xiao, Z. Lin, and M. Xiao, "An integrative framework for precision medication in hypertension: Applying a dialectical perspective to traditional Chinese and Western medicine," *Journal of Holistic Integrative Pharmacy,* vol. 6, no. 3, pp. 290-300, 2025.

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=5824891